AugGen:
Synthetic Augmentation using Diffusion Models Can Improve Recognition

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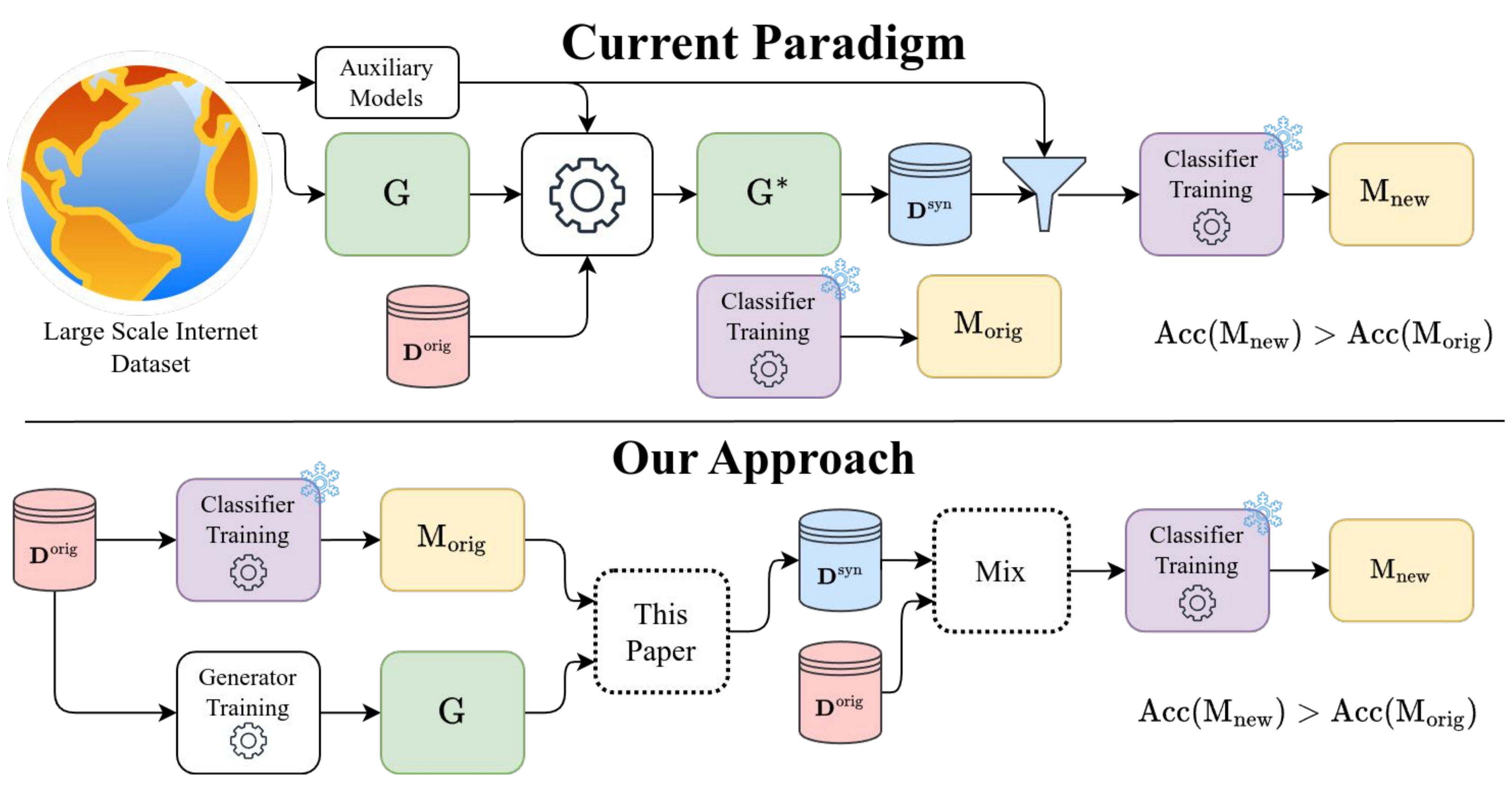
Looking for internship/full-time opportunities

Motivation

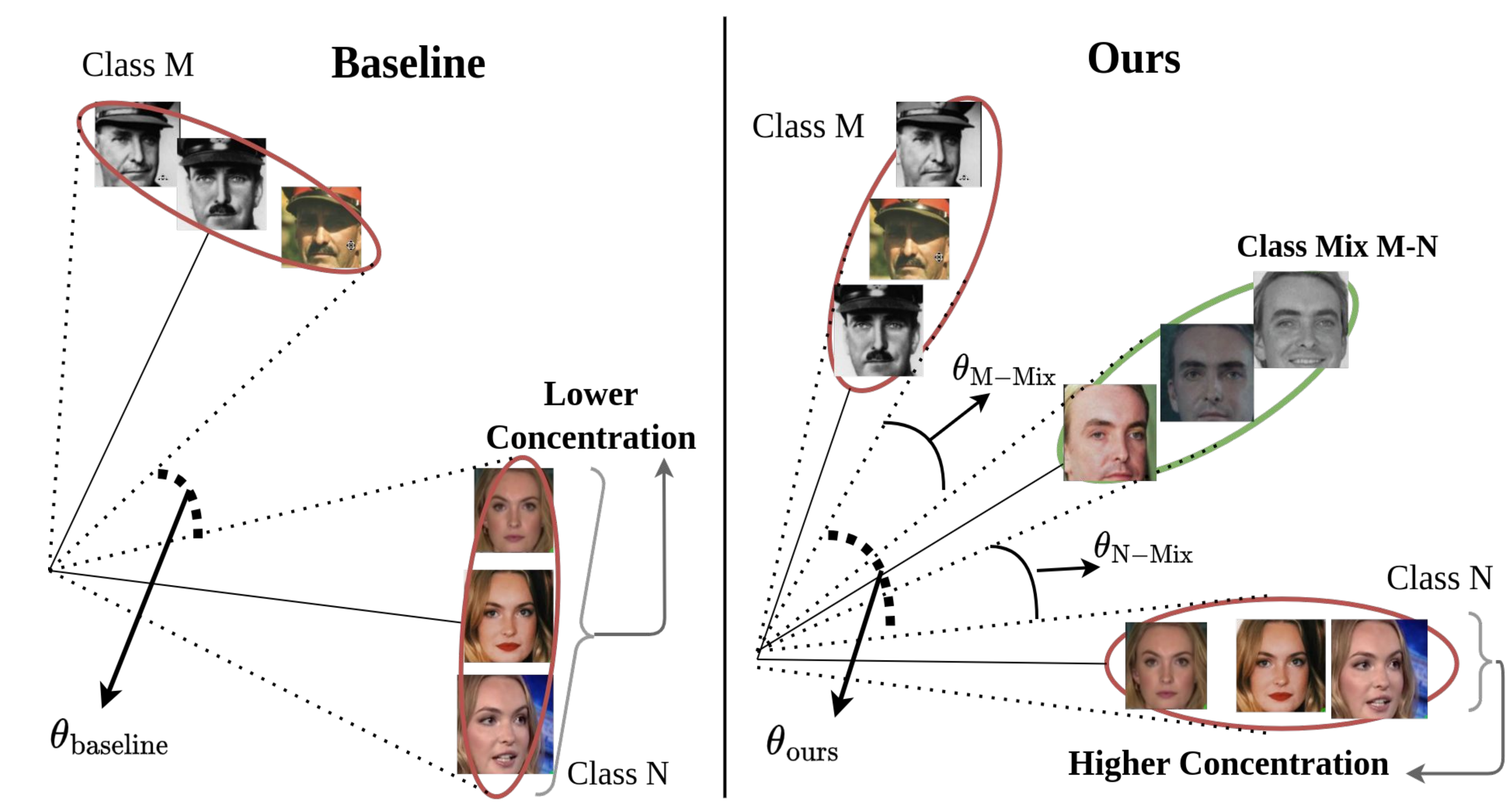
High-quality data for sensitive tasks such as face recognition is **scarce** and subject to **privacy** and **legal** issues.

Synthetic augmentation and **data generation** are a solution, but existing models rely themselves on external datasets!

Can we come up with a **self-contained method for synthetic data generation** that boosts downstream performance?



Idea



TLDR. We use a diffusion generator to sample new images by mixing original classes, creating synthetic identities that stay coherent while lying near the decision boundaries of the original recognition model.

Method

TLDR. We search for conditions (C*) in the generators' condition space that (1) maximises the distance between source classes and (2) minimise intra-class distances.

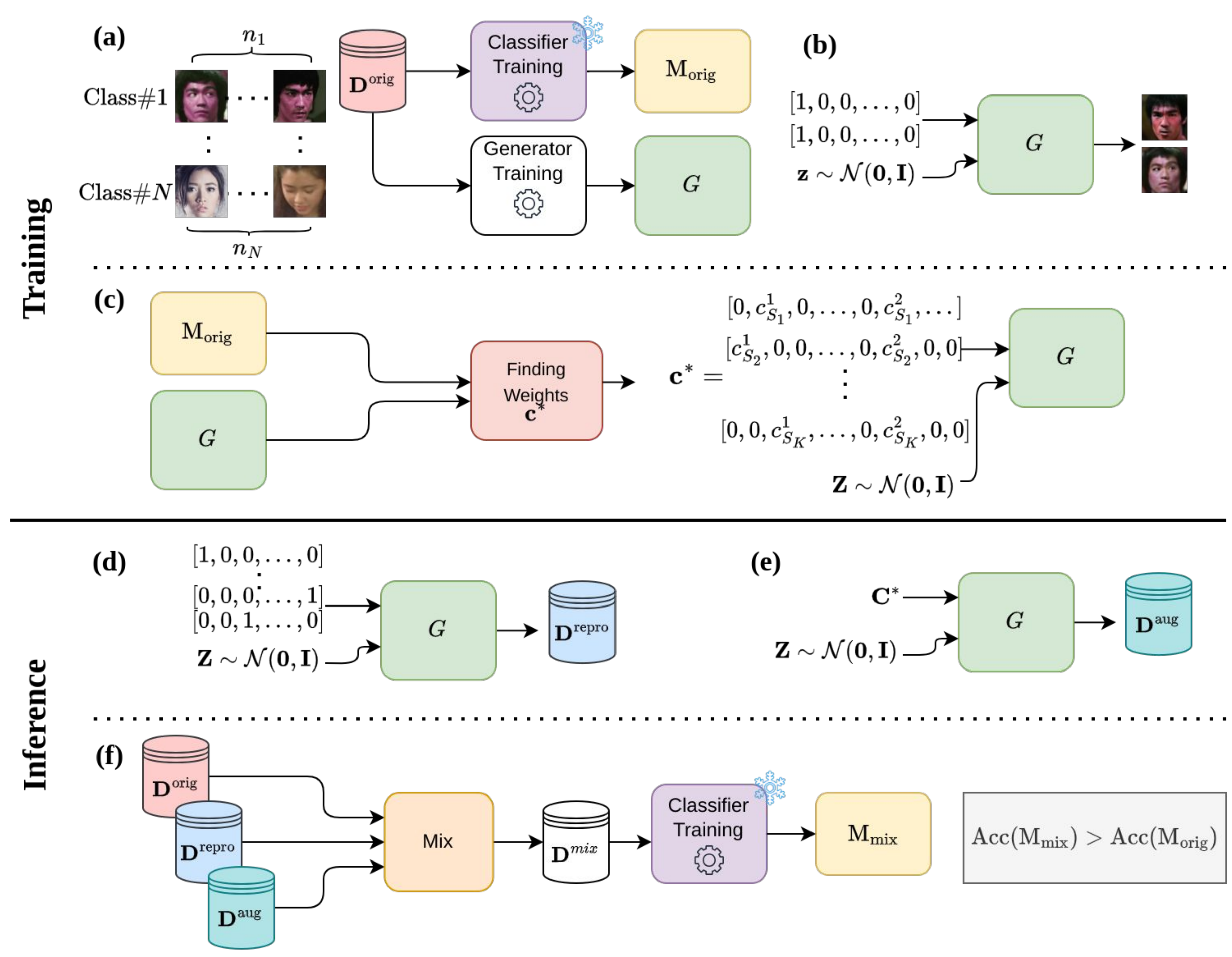
Train class-conditional diffusion generator + SOTA FR model on a labeled dataset.

Search over label/condition mixes in the generator to find settings whose synthetic samples are **consistent** yet **distant** from the source classes.

Why search? optimizing against conditions are difficult in multi step generators like Diffusion models.

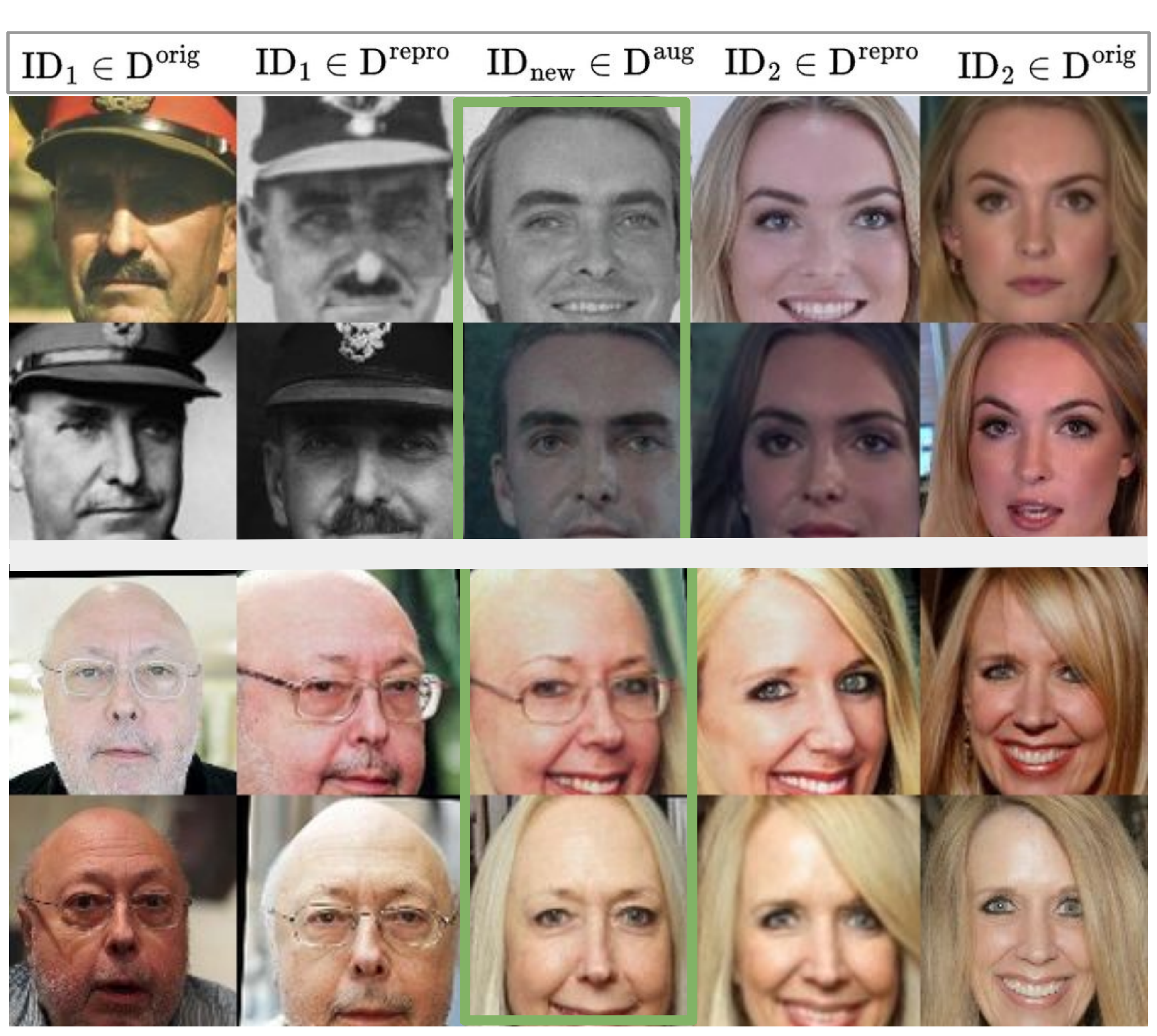
Treat each mix as a new identity, sample multiple images per mix, and retrain FR on real + synthetic data.

Training with the mix yields gains comparable to ~1.7x real data **without any external generator/models besides the original data.**



Samples

AugGen samples (middle) inherit cues from source classes but can be considered as novel identities.



Results

We beat most existing face recognition synthetic dataset generation methods, even though we use less real data.

The improvements often surpass architectural enhancements.

Method/Data	Aux	n^s	n^r	B-1e-6	B-1e-5	C-1e-6	C-1e-5	TR1	Avg-H
CASIA-WebFace	N/A	0	0.5M	1.02±0.26	5.06±1.70	0.73±0.19	5.37±1.41	58.12±0.31	94.21±0.09
CASIA-WebFace †	N/A	0	0.5M	0.74±0.31	3.94±1.62	0.38±0.13	3.92±1.96	59.64±0.49	94.84±0.07
IDiff-face	Y	1.2M	0.5M	0.89±0.07	5.80±0.63	0.70±0.11	7.46±2.08	59.32±0.34	**94.86±0.02**
DCFace	Y	0.5M	0.5M	0.26±0.11	1.59±0.51	0.18±0.07	1.54±0.59	56.60±0.41	94.72±0.09
D^{aug} (Ours)	N	0.6M	0.5M	**2.61±0.91**	**15.74±3.20**	**4.36±1.41**	**18.58±3.99**	**59.82±0.13**	94.66±0.03
WebFace160K	N/A	0	0.16M	32.13±1.87	72.18±0.18	70.37±0.75	78.81±0.32	61.51±0.16	92.50±0.02
WebFace160K †	N/A	0	0.16M	34.84±0.49	74.10±0.24	72.56±0.02	81.26±0.14	62.59±0.01	93.32±0.12
D^{aug} (Ours)	N	0.6M	0.16M	36.62±0.77	78.32±0.33	78.58±0.15	85.02±0.15	61.60±0.38	94.17±0.08

Training a face recognition model on (AugGen samples + original data) yields significant gains, comparable to ~1.7× the amount of real data.

Syn #Class × #Sample	n^r	n^s	B-1e-6	B-1e-5	C-1e-6	C-1e-5	Avg-H	Ratio
0	160K	0	32.13±1.87	72.18±0.18	70.37±0.75	78.81±0.32	92.50±0.02	1
(25K x 20)	160K	500K	36.35±0.70	77.87±0.52	**78.61±0.42**	84.49±0.01	94.10±0.08	1
(30K x 20)	160K	600K	**36.62±0.77**	**78.32±0.33**	78.58±0.15	85.02±0.15	94.17±0.08	1
0	160K + 80K	0	33.78±1.11	77.29±0.12	77.38±0.10	83.50±0.04	93.85±0.02	1.5
0	160K + 110K	0	33.53±1.47	78.26±0.05	78.49±0.54	**85.02±0.01**	**94.19±0.01**	1.69